



VALIDATION REPORT

BLITTA - TOGO SOLAR POWER PLANT



Document Prepared by: LGAI Technological Center, S.A. (Applus+ Certification)

Project Title	BLITTA - TOGO SOLAR POWER PLANT
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Client	ClimatePartner GmbH
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Prepared By	LGAi Technological Center, S.A.(Applus+ Certification)

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Summary:

Validation purpose: The project BLITTA - TOGO SOLAR POWER PLANT undertaken by ClimatePartner GmbH is one of the largest in West Africa and the first renewable energy facility in the country. The plant is which is located in Blitta and the geo-coordinates of the project location are 8° 21'52.33"N, 1° 1'4.42"E. The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by 01-December-2023 bringing the total to 70 MW. This plant supplies reliable, clean electricity to the national grid which is then supplied to hundreds of thousands of homes and businesses in the country and saves CO₂ emissions. Thus, the project has contributed to the reduction of GHG emissions by displacing fossil fuel-dominated grid-based electricity with renewable energy – solar energy based renewable electricity.

The start date of the project activity is the project's COD (Commercial Operation Date) which is 12-June-2021, which is the date on which the project started generating emission reductions. According to the Declaration of Conformity signed by The Minister Delegate to the President of the Republic in charge of Energy and Mines, and the Minister of Economy and Finance, the commissioning date of Phase 1 commenced on 12-June-2021 while the commissioning date of Phase 2 is going to be 01-December-2023. An undertaking has been submitted by PP for double counting confirming that no GHG reduction will be claimed in any other GHG mechanism. Project activity has undergone continued operation and no major breakdown had taken place except for routine maintenance/shutdown.

The project activity, which has a lifetime of 25 years, is expected to produce more than 107,027 MWh of electricity annually. The total estimated emission reductions of the project are 490,607 tCO_{2e} during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid around 61,326 tCO_{2e} annually.

Togo, which imports more than half of its energy from Nigeria and Ghana, is banking on this BLITTA - TOGO SOLAR POWER PLANT project to develop access to electricity for its eight million residents. The Togolese Government understands that the key to economic development and growth depends on access to electricity. The electrification strategy of Togo has set ambitious targets to achieve universal access by 2030.

The project will not only reduce GHG emissions but is expected to achieve many social benefits such as improving social services to targeted social groups and generating job opportunities. Many job opportunities have already been created for the local people that are intended to promote sustained, inclusive, and sustainable economic

growth. The project also helped ensure that the local stakeholders acquire the knowledge and skills needed to promote sustainable development, lifestyles, human rights, and gender equality through education.

The objective of this validation activity is to have an independent third party for the assessment of the project design, and estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project's monitoring plan is assessed against "ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0.
- The project's compliance with the requirements of VCS guideline version 4.2 and standard version 4.3.
- CDM Validation and Verification Standard for project activities, version 03.0

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VCUs).

A risk-based approach has been followed to perform this validation activity. In the course of validation, 14 Corrective Action requests (CAR), 00 Clarification Requests (CLs), and 00 Forward action requests (FARs) were raised and successfully closed. The review of the VCS PD and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews, and PP have provided VVB with sufficient evidence to verify the fulfillment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus + Certification) has been appointed by “ClimatePartner GmbH” to perform the validation of the project entitled “BLITTA - TOGO SOLAR POWER PLANT” under VCS standard V 4.3 and guideline version 4.2. The objective of this validation activity is to have an independent third party for the assessment of the project design, ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable VCS and CDM (when applicable) requirements. In particular;

The project’s baseline is assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 20.0.

The projects compliance with the requirements of VCS guideline V4.2 and standard version V4.3.

CDM Validation and Verification Standard for project activities version 03.0

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified certified units (VCUs).

1.2 Scope and Criteria

The scope of the validation is the independent and objective review of the Project Description. The VCS PD is reviewed against the relevant criteria (see 1.1) and decisions by the CDM Executive Board and VCS Executive Board, including the approved baseline and monitoring methodology. The validation was based on the guidance given in the VCS guideline version 4.2 and standard version 4.3.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can’t be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Level of Assurance

The validation has been planned and organized to achieve a Reasonable Level of assurance as per the requirement of VCS.

1.4 Summary Description of the Project

The project BLITTA - TOGO SOLAR POWER PLANT undertaken by ClimatePartner GmbH is one of the largest in West Africa and the first renewable energy facility in the country. The plant is which is located in Blitta and geo-coordinates of the project location are 8° 21'52.33"N, 1° 1'4.42"E. The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by 2023 bringing the total to 70 MW. This plant supplies reliable, clean electricity to the national grid which is then supplied to hundreds of thousands of homes and businesses in the country and saves CO₂ emissions. Thus, the project has contributed to the reduction of GHG emissions by displacing fossil fuel-dominated grid-based electricity with renewable energy –solar energy based renewable electricity.

The start date of the project activity is the project's COD (Commercial Operation Date) which is 12-June-2021, which is the date on which the project started generating emission reductions. According to the Declaration of Conformity signed by The Minister Delegate to the President of the Republic in charge of Energy and Mines, and the Minister of Economy and Finance, the commissioning date of Phase 1 commenced on 12-June-2021 while the commissioning date of Phase 2 is going to be 01-December-2023.

Table 1 - Specifications of the PV Panels

Manufacturer: Jinko Solar

Power installed on the ground: 50 MWp with 20 MWp planned expansion

Operating Temperature (°C): -40°C~+85 °C

Power Tolerance: 0~+3%

The technical specifications of each type of the modules are shown in the tables below:

Module Type	JKM390M-72-V
Maximum Power (P _{max})	390Wp
Open-circuit Voltage (V _{oc})	49.3V
Maximum Power Current (I _{mp})	9.49A
Module Efficiency STC (%)	19.67%
No of modules	55,242

Module Type	JKM395M-72-V
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Maximum Power (P_{max})	395Wp
Open-circuit Voltage (V_{oc})	49.5V
Maximum Power Current (I_{mp})	9.55A
Module Efficiency STC (%)	19.92%
No of modules	66,300

Module Type	JKM400M-72-V
Maximum Power (P_{max})	400Wp
Open-circuit Voltage (V_{oc})	49.8V
Maximum Power Current (I_{mp})	9.60A
Module Efficiency STC (%)	20.17%
No of modules	5,940

Module Type	LR5-72HBD-545M
Maximum Power (P_{max})	545Wp, Bifacial
Open-circuit Voltage (V_{oc})	49.8V
Maximum Power Current (I_{mp})	13.12A
Module Efficiency STC (%)	21.3%
No of modules	36,708

The Solar PV modules and their specifications mentioned above were crosschecked and confirmed by the assessment team with an amendment to project module supply agreement between AMEA Technical Services LLC and Jinko Solar Co. Ltd. dated 27/02/2020.

Table 2 - Specifications of the Inverters

Make and model	Sungrow SG250HX- 250KTL – H1 String or equivalent
Nominal apparent power	250 kVA @30°C/ 225 kVA @40°C/200 kVA @50°C
Rated current (I_n)	126.3A @40°C, 121.3A @45°C, 108.3A @50°C
Maximum apparent power	200 × 250,000 VA

Rated output voltage	800 V
Type of connection	Triphase
No. of inverters	200

Table 3 -Specifications of the Inverter Flow Transformers

Model	External mineral oil (STS-6000K) or equivalent
Rated power	6 × 6000 kVA @40°C/5400 kVA @50°C and 2 x 8000 kVA @40°C
Primary voltage	800 V
Secondary voltage	33 kV
No-load losses	6.25% (±10%)
Losses at rated current	<4.8 kW (IEC 60076-1) (IEC 62271-202, EN 50588-1, IEC 61429-1)

The investors, the Inverter Flow Transformers and their specifications mentioned above were crosschecked and confirmed by the assessment team with a purchase order placed to Sungrow Power Supply Co. Ltd. dated 08/01/2020.

The project activity, which has a lifetime of 25 years, is expected to produce more than 107,027 MWh of electricity annually. The total estimated emission reductions of the project are, 490,607 tCO_{2e} during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid around 61,326 tCO_{2e}, annually.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation Scope: The scope is defined as an independent and objective review of the project design document. The VCS PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard version 4.3 and guideline version 4.2, including the approved baseline and monitoring methodology “ACM0002 - Large-scale Consolidated Methodology, Grid-connected electricity generation from renewable sources (Version 20.0)”. The validation was based on the requirements in the Validation and Verification Standard for project activities version 03.0, and VCS guideline version 4.2 and standard version 4.3.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project document.

Validation Process: The project assessment is based on the Clean Development Mechanism Validation and Verification Standard for project activities version 03.0 and VCS standard version 4.3 and guideline version 4.2 and is conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

- Once the project is received by the assessment team, the members of the assessment team carried out:
- A desk review of the Project Design Document;
- Follow-up interviews with project stakeholders;
- The resolution of outstanding issues and the issuance of the final Validation report and opinion.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI TECHNOLOGICAL CENTER S.A. (APPLUS+ CERTIFICATION) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from validating the identified criteria.

Appointment of the assessment team:

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A).
- Technical Expert (TE).
- Technical Reviewer (TR).
- Any of the above mentioned roles in training (iT, e.g. AiT for auditor in training).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the Assessment Team as shown below:

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Mr. Pankaj Kumar	LA/TE	YES	YES	YES	YES
Mr. Srikanth Meesa	TR	YES	YES	YES	NA

The complete list of CVs is included as Appendix 3 of this report.

Document Review:

The VCS PD submitted by the Client was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

Follow -up Interviews:

A remote audit is conducted by LGAI Technological Center S.A. (Applus+ Certification) assessment team who performed interviews, telephone conferences, and remote inspection through ICT means with project stakeholders to confirm selected information and to resolve issues identified in the document review. The detail is provided in this report in the below sections.

Resolution of Clarification and Correction Action Request:

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the project design. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and Applus+ Certifications to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 2.

The VCS PD Version 1 submitted by project owners on 17-May-2022 serves as the basis for the final assessment presented. Additional changes to the project during the validation process is not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality Control:

As final step of a validation, the final documentation including the final validation and the checklist may undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of interest.

After confirmation of the project owners, the positive validation opinion and relevant documents are submitted to the project proponent or its representatives for the submission to the VCS secretariat through the VCS web platform.

2.2 Document Review

As a part of audit, the assessment team conducted validation to confirm the project description based on the supporting documents submitted and to assess whether they are as per the state requirements.

The PDD submitted by the PP to Applus+ Certification. This document was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done.

The main objectives of the desk review step as part of the validation assessment are summarized as follows: -

- Verifying the completeness of the data and the information presented in the initial version of the PDD.
- Checking the compliance of the PDD with respect to the monitoring plan described in the approved methodology for the VCS project activity, inter alia by verifying that all applicable monitoring requirements (as established by the applied CDM baseline and monitoring methodology) were sufficiently and reasonably met for the project activity. Particular attention to the frequency of measurements, specification requirements and functional aspects of monitoring equipment/instruments was paid by the Applus+ Certification's validation team (including compliance of applicable calibration and functional requirements of equipment/instruments and the quality assurance and quality control procedures considered as part of the application of the monitoring plan of the VCS project activity).
- Evaluating data management and the quality assurance and quality control system in the context of their influence on fairly generation and appropriate reporting of the operation of the VCS project activity as well as reporting of estimated emission reductions.

A complete list of documents reviewed by the Applus+ Certification's verification team is available in Appendix 1 of this Verification Report.

2.3 Interviews

Remote audit was conducted for the project activity on 12-August-2022. The VVB has taken alternative measures to reach reasonable level of assurance and conducted remote audit with site personal & consultant and local stakeholders along with the PP representative. Technical details & metering/monitoring arrangement verified through photographs/name plates and calibration certificates shared by PP. All the documents were cross checked to ensure conservative estimation of emission reduction.

During the remote audit, the PP representatives were questioned about the implementation of the project activity. Several topics like the verification of commissioning date of meters, the generation, recording, and monitoring of the data and the error accountability were discussed. To cross check the information provided by PP, various documents like technical specifications, commissioning certificates, Power Purchase agreements, MoU with AMEA SOLAR TOGO SAU and Togolese Govt, calibration certificates, etc. were also verified.

As a part of audit, the assessment team conducted interview of the following stakeholders. The identified personnel, their roles who were interviewed during remote audit who provided information additional to that provided in the project description and any supporting documents is given below;

Date of Interview: 12-August-2022		
Sr. No.	Name of Persons	Role/Designation
1.	Raoug Damtare	QHSE Manager, AMEA SOLAR TOGO SAU
2.	Belzra Kossi	Police
3.	Bierdey Abdel-Ade	Site Engineer, AMEA SOLAR TOGO SAU
4.	Sambanne	Blitta Village (Stakeholder)
5.	Bilake	Blitta Village (Stakeholder)
6.	Herve Rioux	Site Representative, AMEA SOLAR TOGO SAU
7.	Amer Arafat	Consultant
8.	Priscilla Nunnari	Consultant

2.4 Site Inspections

No physical on-site inspection (with the physical presence of the Applus+ Certification's validation team) was conducted as part of the performed VCS validation assessment.

As a result of raised travelling restrictions related to the COVID-19 pandemic, the Applus+ Certification's validation team proposed to the VCS project proponent AMEA SOLAR TOGO SAU, as an alternative, consider postponing such on-site visit by taking into account not only travelling restriction related official decisions and recommendations from local authorities (i.e. restrictions and recommendations from the Government of Togo), but also related travelling restriction policy announced by Applus+ Certification's central office due to the COVID-19 pandemic. As an answer to such proposal from Applus+ Certification, the representatives of the project proponents highlighted to the Applus+ Certification's validation team that they were not in a position to accept any postponing of on-site visit that would result on potential delay on issuance of VCUs for the proposed project activity

Furthermore, the above summarized travelling restrictions were expected to indeterminately continue in light of the COVID-19 pandemic situation in Africa. The content of the VCS Guidance titled "COVID-19 Travel Guidance for Projects issued by VERRA/VCS" (issued on 18/03/2020) was also considered by the Applus+ Certification's verification team.

By acknowledging that the previously planned physical on-site inspection could not be performed as part of the validation assessment due to related travelling restrictions because of the COVID-19 pandemic, by also assuming that such on-site inspection could not be postponed, the Applus+ Certification's validation team thus performed its document review and interviews with representatives of the project proponents (of which details are included in Sections 2.2 and 2.3 respectively) by incorporating the following additional checking's/assessments (complementary auditing measures) as part of a remote inspection to the project site which was performed on 12-August-2022 as per the prepared document VCS – Verification Audit Plan.

Upon previous request from the Applus+ Certification's validation team, the representatives of the project proponent organized the production of a sufficiently complete set of sequential live videos (movies) + photographs (pictures) produced/generated online in the project site by using appropriate ICT tools (e.g. mobile phone with camera) with the goal of making it possible to the Applus+ Certification's verification team to remotely assess and confirm the implementation and operation of the VCS project activity (as if the validation team were on-site).

By watching/visualizing (more than once) the content of the set of sequential live videos (movies) + photographs (pictures), the Applus+ Certification's verification team was able to have a comprehensive and complete remote assessment of the VCS project activity as if the team were on-site.

Details about all monitoring instruments/equipment (of which the latest version of the PDD refers to).

Details about the functioning of the process for gathering, recording and archiving of monitoring data (i.e. measurement records for electricity generated by the VCS project activity).

Watching and visualizing the set of sequential live videos (movies) + photographs (pictures) sufficiently the following assessment outcomes: -

The possibility of having a complete and transparent remote visualization of the implementation and operation of available information and data flows/procedures for generating, gathering, aggregation, recording and reporting data for the ex-post determined monitoring parameter (as if the Applus+ Certification's verification team were on site), thus making it possible to the verification team to confirm the correctness and appropriateness of related data gathering, processing, recording and data recording by the available project's monitoring infrastructure as well as to confirm correctness of related information included in the PDD for the VCS project activity .

The possibility of performing a remote (but comprehensive) checking of all project's monitoring instruments/equipment (including confirmation of occurred performance of appropriated related calibration events in such instruments/equipment) as well as checking of monitoring practices as part of the operation of the VCS project activity vis-à-vis related requirements of the applied methodology. Such checking included confirmation of existence and archiving in the project site of original documents (e.g. certificates of calibration) in the project site + additional documentation used for cross-checking of calculation and information, with electronic format copies of such documents and evidences being also made available to the Applus+ Certification's verification team for its remote assessment (as part of the performed desk review assessment phase).

The possibility of performing a comprehensive checking of the project's quality control and quality assurance (QA/QC) procedures in place to prevent and/or identify and/or correct errors or omissions in the reported monitoring parameters.

Based on its accumulated expertise and experience with CDM/VCS assessments for other similar project-based initiatives (including previous experience of the Applus+ verification team with the performance of remote inspections to the project sites of CDM/VCS project activities which promote renewable electricity generation), it is Applus+ Certification's opinion that objectives to be expected for a physical on-site inspection to the project site were sufficiently reached by the Applus+ Certification's verification team through watching/visualizing online of the set of sequential live videos and pictures produced on-site by the representatives of the project proponent on 12-August-2022.

In summary, by taking all above-presented aspects into account vis-à-vis applicable requirements established in VCS Standard (version 4.4) , CDM Validation and Verification Standard for project activities (version 03.0) , VCS/VERRA COVID-19 Travel Guidance for Projects (18/03/2020) and VCS Validation and Verification Manual (version 3.2), as well as the recently published guidelines from the CDM titled "Amendments to version 03.0 of the CDM validation and verification standard for project activities on remote validation or verification by DOEs" (version 01.0), the Applus+ Certification's verification team judged that performing the above-described additional checking's/assessments (complementary auditing measures) instead of performing the previously scheduled physical on-site inspection to the project site is deemed acceptable and

sufficient to have the overall quality and completeness of the performed verification assessment not being negatively affected. It is the opinion of Applus+ Certification verification team that the decision made of not performing a physical on-site inspection as part of the verification assessment and applying the above-summarized additional checking's/assessments (complementary auditing measures) instead are under conformance with applicable guidelines added in the guidance titled VERRA/VCS COVID-19 Travel Guidance for Projects(18/03/2020) which is made available at the VERRA/VCS website

Duration of remote audit: 12-August-2022 (remote assessment)				
No.	Activity performed on-site	Site location	Date	Team member
1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project, Additionality, start date, project ownership and local stakeholder consultation and Monitoring.	Remote Assessment via video conference (Skype)	12-August-2022	Mr. Pankaj Kumar (Team Leader)

2.5 Resolution of Findings

The objective of this validation is to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design and emission reduction report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in Appendix 2.

The final VCS PD Version 07 and ER sheet version 07 submitted by project owners on 11-April-2023 serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project design document	00	01	00
Description of project activity	00	01	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	02	00
- Deviation from methodology	00	00	00
- Clarification on the applicability of methodology, tool, and/or standardized baseline	00	02	00
- Demonstration of additionality	00	03	00
- Emission reductions	00	02	00
- Monitoring plan	00	00	00
- Stakeholder's consultation process	00	01	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting- for validation	00	01	00
Others (please specify)- SDG Goals	00	01	00
Total (Validation)	00	14	00

The list of findings and their resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

This is validation of the project activity and no FAR was raised and during this validation process. Please refer Appendix 2 for details.

3 VALIDATION FINDINGS

3.1 Project Details

BLITTA - TOGO SOLAR POWER PLANT is one of the largest in West Africa and the first renewable energy facility in the country. This is crosschecked and confirmed by the assessment team with the Press Release¹ dated 22/06/2021. This plant, which is located in Blitta, supplies reliable, clean electricity to hundreds of thousands of homes and businesses in the country and will save more than one million tonnes of CO₂ emissions through its lifetime.

¹ <https://www.irena.org/news/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility>

The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by 1st December 2023 bringing the total to 70 MW.

The baseline scenario is the electricity generated and delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The project activity, which has a lifetime of 25 years, is expected to produce more than 107,027 MWh of electricity annually. The total estimated emission reductions of the project are 490,607 tCO_{2e} during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid around 61,326 tCO_{2e} annually.

The coordinates of the project location crosschecked by the assessment team via google maps and file shared by the PP and are 8°21'52.33"N, 1°1'4.42"E. The energy generated by this project, is supplied to the national grid through La Compagnie Energie Electrique du Togo.

According to the latest VCS Standard, version 4.3, the project is eligible under the scope of the VCS Program because:

- It results in CO₂ emission reductions, one of the seven greenhouse gases under the Kyoto Protocol
- It consists of grid-connected electricity generation using (large scale) solar power plant built in a Least Developed Country.

The project has been designed to include a single installation of an activity and it includes a single location or installation only.

The validation team has checked and found that the proposed project is not a grouped project.

The validation team checked remotely and confirms that the details of the project proponent areas below:

Organization name	AMEA SOLAR TOGO SAU
Contact person	Basma Bentaher
Title	Senior Director – Project Development
Address	Marina Plaza Offices 3301/3302/3303 Level 33 Dubai Marina PO Box 37669 Dubai United Arab Emirates
Telephone	+971 4 310 7014
Email	Basma.Bentaher@ameapower.com

Organization name	ClimatePartner GmbH
Contact person	Dr. Sascha Lafeld
Title	Head of Carbon Offsets and Green Energy Services
Address	St.-Martin-Str. 59, D-81669 Munich, Germany
Telephone	+49 89 1222 8750
Email	sascha.lafeld@climatepartner.com

Other entity involved is Element 6 – FZCO as project consultant and details are as below:

Organization name	Element 6 – FZCO
Role in the project	Consultant
Contact person	Amer Arafat
Title	Partner
Address	Technohub 1, DTEC, Dubai Silicon Oasis, Dubai, UAE
Telephone	+971 52 938 9590
Email	amer@element6.cc

Project Start Date:

The project's COD (Commercial Operation Date) is 12-June-2021, which is the date on which the project started generating emission reductions. According to the Declaration of Conformity signed by The Minister Delegate to the President of the Republic in charge of Energy and Mines, and the Minister of Economy and Finance. The validation team checked the commissioning details from the commissioning certificates and the declaration of conformity provided by the project proponent and start date found correct.

Project Crediting Period:

The validation team confirms that the crediting period dates for the project is as below:

The project's crediting period is 7 years renewable.

The first crediting period starts from 12-June-2021 and lasts till 11-June-2028

Project scale & estimated GHG Emissions Reductions or Removals:

The validation team confirms that the project involves setting up single installation of a project activity project with large scale SPVs. The installed capacity of the project is 70 MW. The estimated annual GHG emission reductions/removals of the project are under 20,000 – 100,000 tCO_{2e}/year.

As per the section 3.9.1 of VCS standard version 4.3, the projects are classified as follows:

1. Projects: Less than or equal to 300,000 tonnes of CO_{2e} per year
2. Large Projects: Greater than 300,000 tonnes of CO_{2e} per year

Hence, the proposed activity instances are classified as “Project”.

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO _{2e})
2021	39,985
2022	70,822
2023	70,468
2024	70,308
2025	69,765
2026	69,416
2027	69,069
2028	30,774
Total estimated ERs	490,607
Total number of crediting years	7
Average annual ERs	61,326

Conditions Prior to Project Initiation:

The validation team during the desk review and remote assessment confirms that in accordance with methodology ACM0002, Version 20 the project activity is the installation of a new grid-connected renewable power plant (Greenfield). The baseline scenario is defined as electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system” version 7.0

Project compliance with applicable laws, statutes and other regulatory framework:

The validation team confirms that the Project has received necessary approvals for development and commissioning for the project activity. The Togolese government created the Togolese Rural Electrification and Renewable Energies Agency (AT2ER) (Presidential decree N ° 2016 - 064 / PR of May 11th, 2016)², whose mission is to promote renewable energy in Togo. The creation of the AT2ER was accompanied in July 2018 by the announcement of the Law 2018-010 08/08/2018 on the promotion of electricity production based on renewable energy sources in Togo. This resulted in the creation of a specific legal framework to respond adequately to the challenges related to the production, marketing and use of electricity generated from renewable energy in Togo.

Project Ownership:

AMEA SOLAR TOGO SAU is project proponent (PP) of project activity and they have the legal right to control and operate the project activities.

The project ownership has been checked by the validation team and demonstrated through below supporting documents:

- Certificate of Completion signed by the Togolese Republic
- Power Purchase Agreement (PPA) signed by AMEA SOLAR TOGO SAU and the Togolese Republic
- Declaration of Conformity signed by the Togolese Republic
- Entry into Force of the Concession Agreement signed by AMEA SOLAR TOGO SAU and the Togolese Republic
- Back Energization of the PV Plant Document signed by the Togolese Republic, the Electricity Off-taker and AMEA SOLAR TOGO SAU.

² <https://www.devex.com/organizations/togolese-rural-electrification-and-renewable-energies-agency-at2er-136414>

Participation Under Other GHG Programs:

The validation team found the declaration regarding the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits submitted by project participant correct and confirms that, the project activity is not involved in other Emissions trading programs and other binding limits.

Leakage Management for AFOLU project:

Not applicable to the project activity.

Commercially Sensitive Information:

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the validation team for analysis which lead to positive conclusion for this validation.

Sustainable Development:

According to World Bank data from 2020, around 54% of the Togolese population had access to electricity. In order to meet the needs of the Togolese people, the Government of Togo has committed to an ambitious policy on energy access that is to achieve a 100% electricity access rate in the country by 2030³. In this context, the country has subscribed at regional and international level to many commitments in this sector and which require that the share of renewable energy to be increased to 50% in the national energy mix by 2025⁴. This is in accordance with Goal 7 of the United Nations Sustainable Development Goals, which aims to ensure access for all to reliable, sustainable, and modern energy services at affordable costs.

In addition, AMEA Solar Togo SAU has put in place several CSR activities across essential sectors. Further, in line with VCS Standard Version 4.3, para 3.16.1, "The project proponent must demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period, and in each subsequent monitoring period." As per the provision of the implementation deadline, "Projects registered before 20-January-2023 shall demonstrate contributions to at least three SDGs by 20-January-2025". However, PP has provided information on the SDG contributions by the project activity for SDGs. The activities are developed in consultation with

³ https://unfccc.int/sites/default/files/NDC/2022-06/CDN%20Revis%C3%A9es_Togo_Document%20int%C3%A9rimaire_rv_11%2010%2021.pdf

⁴ https://unfccc.int/sites/default/files/NDC/2022-06/CDN%20Revis%C3%A9es_Togo_Document%20int%C3%A9rimaire_rv_11%2010%2021.pdf

the community near the BLITTA - TOGO SOLAR POWER PLANT and are in accordance with the following Sustainable Development Goals (SDGs):

SDG	SDG Overview	Action Taken	Monitoring Parameters
3 Good Health and Wellbeing	Ensure healthy lives and promote well-being for all at all ages	- Construction of a clinic in the village of Blitta for curative and preventive service to the members of the local community. - Distributed essential food and hygiene products to 2,200 families as a response to the lockdown in Togo.	3.1. Number of patients visiting the newly constructed clinic (700 patients per year)
4 Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	- Construction of three primary schools (Blitta, Tomegbe and Kokonade) - Renovation of four primary schools (Blitta Centre, Carrefour, Finade and Tchangaide). - Provision of necessary furniture and educational equipment - Internship program for 36 students from Lomé Technical University	4.1. Number of students going to primary schools (750 students per year)
7 Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable, and modern energy for all	Construction and commissioning of the BLITTA - TOGO SOLAR POWER PLANT	7.1. Total clean energy generated from the BLITTA - TOGO SOLAR POWER PLANT. Measured in MWh/year. (Around 88,000 MWh for 2022, increasing to around 99,000 MWh in 2023 and around 120,000 MWh in 2024 onwards)

8 Decent Work and Economic Growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment, and decent work for all	Temporary jobs created during the construction of the plant and permanent jobs created for the operation and maintenance of the plant	8.1. Number of employees working in the operation and maintenance activities of the plant (10 employees)
13 Climate Action	Take urgent action to combat climate change and its impacts	Reduction of total carbon dioxide emissions generated.	13.1. Amount of annual emission reductions achieved from the project activity. Measured in (tCO ₂ e) ((Around 50,400 tCO ₂ e for 2022, increasing to around 56,700 tCO ₂ e in 2023 and around 68,700 tCO ₂ e MWh in 2024 onwards)

The validation team has assessed the SDG 3, SDG 4, SDG 7, SDG 8 and SDG 13 and supporting documents shared by PP and found values correct as provided in Section 1.17 of VCS PD version 07.

Hence, validation team confirms that the purpose of this project activity is to reduce GHG emissions by installing Solar Photovoltaic (SPV) Panels by *AMEA SOLAR TOGO SAU*. The same is confirmed through document review and remote interviews with the project site personnel. Final PD also reflects the same. Thus, found acceptable as PD is accurate and provided complete project information.

3.2 Safeguards

3.2.1 No Net Harm

The activities during construction, operation, and decommissioning of the project have been analysed to identify corresponding environmental and social impacts and assess their significance. Mitigation measures have been identified in the Environmental and Social Management Plan (ESMP) for negative impacts where institutional responsibilities for implementation and follow-ups have been presented. The socioeconomic benefits of the project, therefore, outweigh the potential negative environmental and social impacts.

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The stakeholder meeting took place at Centre International de Conférence de Blitta (CICB) on Tuesday, 05-July-2022 at 08.00 AM GMT.

The objective of the Stakeholders' Consultation event was to take due diligence of all comments, questions, and opinions in an open and transparent manner towards the projects' economic, social, and environmental impacts, and sustainability. It was also to raise the stakeholders' awareness level about the procedure to register such projects under the VCS.

The Local Stakeholder Consultation report is devised to provide the outcomes of the stakeholder consultation and to ensure the transparency of the process through documentation.

The invitees were sent the invitation to attend the stakeholders' consultation event. They were also requested to submit their written comments, should they have any, via email to Mr. Amer Arafat at info@element6.cc.

The validation team verified the documents submitted by the PP like the attendance sheet, invitation letter, and feedback provided by the local stakeholder. Thus, the validation team is of opinion that the local stakeholder consultation is adequate and appropriate.

The Validation team has also confirmed with the PP that a grievance mechanism is in place consisting on the appointment of a Community Liaison officer in charge of managing community complaints and periodic awareness campaigns for the local populations. The ongoing communication mechanism, as described in the VCS PD Section 2.2., has been included as a topic on the LSC process and this has been found acceptable.

3.2.3 Environmental Impact

The project activity is having positive impacts and no significant adverse environmental impacts are foreseen. Since, the project activity is an electricity generation from renewable source (i.e., solar energy) therefore no negative impacts are envisaged.

An Environmental and Social Impact Assessment (ESIA) was carried out for the project. The activities during the construction and operation of the project have been analysed to identify correspondent environmental impacts and assess their significance. The summary of the findings of the report were as follows:

Impact Assessment methodology

The following assessment of impacts distinguishes between the construction, operation, and decommissioning phases.

- Land use
- Biodiversity
- Air quality and noise levels
- Water resources
- Traffic and Transport
- Existing infrastructure
- Visual impacts
- Socioeconomic impacts

Land, land use, and waste management

As the land consists of trees and agricultural activities, there might be some changes in the soil properties resulting from the soil replacement for the foundations and from construction of the plant intermediate road. Such impacts could only affect the subsequent use of the project site after decommissioning and would be mitigated through the site preparation works. There might also be some impacts on the soil at the project site resulting from accidental leaks of wastes or oils used by the construction machinery. However, such spills are expected to be limited through the Spill Prevention Plans and having spill kits available throughout the site and the Hazardous Waste Management plan.

Impacts on land beyond the footprint of the project will also include domestic solid waste in addition to some construction wastes such as plastics, metals, wood, etc. Because the quantity of such wastes is expected to be limited, the correspondent impacts on waste disposal sites are minor.

Air quality

The air emissions expected during the construction of the project were the airborne dust resulting from earthworks, in addition to the emissions of different construction machinery such as power generators, excavators, loaders, vehicles, etc. In the absence of sensitive receptors in the area, emissions during construction, such as exhaust gases have low significant impact, and may only temporarily affect users, as the average speed on the road is quite high, during passing the area in addition to slight impacts on the construction workers.

Mitigation measures such as spraying loose soil during excavation and filling operation can be adopted to minimize airborne dust.

Noise levels

Normally construction works include noisy activities related to the operation of construction machines, possible hammering, drilling works, soil compaction in addition to the noise generated from construction related trucks. The impacts of construction noise are likely to cause some considerable effects on neighbors as the neighboring areas are closely populated.

The contractor was encouraged to minimize noise emissions, such as turning off engines of unused equipment, use lower noise emission equipment to the extent possible and use silencer for noisy equipment (such as generators) to comply with the local noise regulations.

Water resources

During construction, temporary movable construction yard (for storage of materials and servicing of machinery) and temporary offices, for construction staff, were erected at a place. The area included simple sanitary facilities. Water supply was via tankers from the central pipeline, which will normally cover domestic uses, uses for soil spraying and uses for mixing concrete. The amount of water consumed in construction sites are expected to be relatively little amounts, in which the quantitative impacts on water resources are negligible. The mitigation measures include collecting wastewater in waterproof collection tanks and evacuate it frequently to the nearest wastewater treatment plant or sewerage system.

Traffic and Transport

During the construction, heavy transport was used to transfer PV modules and other electric equipment to the construction sites. Since the road networks have low levels of service, truck convoys carrying equipment to the site may cause bottlenecks and impact the current traffic flow.

A plan to develop traffic management was introduced to ensure equipment is delivered in a timely fashion and in good condition, whilst complying with the local legislation.

3.2.4 Public Comments

The validation team noted that this project was open for public comment from 29-August-2022 to 28-September-2022. No comments were received.

The details were checked by the validation team in the following web platform⁵.

3.2.5 AFOLU-Specific Safeguards

Since it is a non-AFOLU project, this section is not required.

⁵ <https://registry.terra.org/app/projectDetail/VCS/3298>

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Methodology: ACM0002: Grid-connected electricity generation from renewable sources --- Version 20

Project Type: Renewable energy projects (Large Scale)

Sectoral Scope: 1

The methodology refers to the following methodological tools that have been considered by the project):

TOOL01: Tool for the demonstration and assessment of additionality, version 07.0

TOOL07: Tool to calculate the emission factor for an electricity system, version 07.0

Tool 24: Common practice analysis, version 3.1

Tool 27: Investment analysis, version 9.0

3.3.2 Applicability

The applied baseline methodology is justified as it has been demonstrated that the proposed project activity is:

For ACM0002: Grid-connected electricity generation from renewable sources --- Version 20

Methodological Applicability	Project Justification	VVB Assessment
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units;	Option (a) - The project is a grid-connected solar energy power generation project that is a greenfield installation.	The project activity a new grid connected renewable energy power plant (Solar) at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable. Validation team checked O & M contract and PO to

(d) Involve a rehabilitation of (an) existing plant(s)/unit(s);or (e) Involve a replacement of (an) existing plant(s)/unit(s).		confirm that the plant is a greenfield renewable energy power plant.
The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydropower plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been	Option (a) – The project activity is a renewable energy (solar power) power plant.	The project activity a new grid connected renewable energy power plant (Solar) at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable. Validation team checked O & M contract and PO to confirm that the plant is a greenfield renewable energy power plant. The project activity does not involve any capacity additions, retrofits or replacements and therefore this second condition is not applicable.

undertaken between the start of this minimum historical reference period and the implementation of the project activity.		
In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4W/m², all of the following conditions shall apply:	Not applicable. This project is not a hydro power plant.	The proposed project activity is a grid connected renewable energy project (Solar). This condition is applicable only for hydro power plants and not applicable for solar projects. Hence this criteria is not applicable.

<i>i.</i> The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; <i>ii.</i> Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; <i>iii.</i> Installed capacity of the power plant(s) with power density lower than or equal to 4W/m² shall be: <i>a)</i> Lower than or equal to 15MW; and <i>b)</i> Less than 10 per cent of the total installed capacity of integrated hydro power project.		
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In the case of integrated hydro power projects, project proponent shall: - Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or - Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Not applicable. This project is not an integrated hydro power plant.	The project activity is a grid connected renewable energy project (Solar). This condition is applicable only for hydro power plants and not applicable for solar power projects. Hence this criteria is not applicable.
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The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	Not applicable. This project activity does not involve switching of fossil fuels to renewable energy or biomass fired power plants.	The project activity is installation of a new grid connected renewable energy project (Solar) and does not involve switching from fossil fuel to renewable energy and hence this criterion is not relevant. The project activity does not involve any biomass-based power plants and hence this criterion is not applicable to the project activity.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable. This project activity is not a retrofit, rehabilitation, replacement, or capacity addition.	The project activity a new grid connected renewable energy plant (Solar) and not a retrofits, replacement or capacity additions and therefore this criterion is not applicable to the project activity.
In addition, the applicability conditions included in the tools referred to below apply.	See below tables.	

TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0 is applicable under the following conditions:

Tool Applicability	Project Justification
The use of the “Tool for the demonstration and assessment of additionality” is not mandatory	Not applicable.

for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	No new methodology is proposed.
Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The additionality tool is applied, hence applicable.

TOOL7 “Tool to calculate the emission factor for an electricity system” version 07.0 is applicable under the following conditions:

Tool Applicability	Project Justification
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project is a grid-connected solar energy power generation project and thus substitutes grid electricity.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW)	Steps involved in calculation of Emission Factor is included in section 3.1 of the VCS PD as per the required tool.

should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The Project is located in non-Annex I country and hence the tool is applicable.
Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not applicable. No biofuel is involved in this project activity.

TOOL24 "Common practice" Version 3.1 is applicable under the following conditions:

Tool applicability	Project Justification
This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	PP has applied "Tool for the demonstration and assessment of additionality". Hence, the current tool is applicable.
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

TOOL27 "Investment analysis" Version 09 is applicable under the following conditions:

Tool applicability	Project Justification
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This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	“Tool for the demonstration and assessment of additionality” has been applied. Hence, the tool is applicable.
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

LGA Technological Center S.A. (Applus+ Certification) confirms that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology is applicable to the project activity.

The project activity qualifies as Type I during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, the total installed capacity of project activity is 70 MW which is applicable as per large-scale project activities methodologies.

CAR 02 was raised on this section and closed successfully. Please refer Appendix 1 for further details.

3.3.3 Project Boundary

As per methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 20

“The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the solar power plant is connected to”.

The electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the solar power plant and that are covered by either single or layered dispatch area.

Thus, the project boundary for this activity is the national/sub-national grid in Togo.

The sources and GHG gases involved for the Project activity are as below:

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Included	Main emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source
Project	CO ₂	Excluded	Minor emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source

3.3.4 Baseline Scenario

According to the ACM0002: Grid-connected electricity generation from renewable sources — Version 20

“If the project activity is the installation of a Greenfield power plant, the baseline scenario is the electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”, version 07.0.

The project activity is the installation of a new power plant at the site and is not a capacity addition or retrofit or replacement of any other existing plant. Any electricity generated and exported from the project activity would in its absence have been generated in the grid dominated by fossil fuel-based power plants, which represents the baseline scenario. The project scenario is a Solar photovoltaic power plant that would generate energy and supply it to the national grid through La Compagnie Energie Electrique du Togo⁶.

⁶<https://energie.gouv.tg/projet/construction-dune-centrale-solaire-de-50-mwc-a-blitta/>

The baseline, the scenario existing prior to the implementation of the project activity, is the continuation of the current practice. The electricity generated by the project activity would have been generated by the addition of new fossil fuel-based generation sources to the grid to supply the additional electricity capacity required.

The validation team has accessed and confirm that PP has identified baseline scenario in line with Tool07 and the assessment on the same is presented below in section 3.3.6. It was confirmed that Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool is applicable for the current project and is correctly applied.

The validation team conclude that, the above-mentioned baseline scenario is in line with the applied methodology “ACM0002: Grid-connected electricity generation from renewable sources --- Version 20”. Same is also reflected in final PD version 07. Thus accepted.

3.3.5 Additionality

The project activity generates renewable electricity by means of solar PV technology, which is a renewable, zero emission source of energy. Additionality of the project activity is demonstrated through a project specific additionality test. The project is not mandated by any law, statute, or other regulatory framework and the Project Proponent has stated the same in Section 3.5. of the VCS PD, which has been confirmed by the assessment team.

Baseline considerations for the project are based on approved consolidated baseline methodology ACM0002 (Version 20.0). The methodology requires the project proponent to determine the additionality based on the “CDM - Methodological Tool “Tool for the demonstration and assessment of additionality”, version 07.0. The tool provides for a stepwise approach to demonstrate and assess the additionality of a project. These steps are: (a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind; (b) Step 1 Identification of alternatives to the project activity; (c) Step 2 Investment analysis; and (e) Step 4 Common practice analysis.

The stepwise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the project activity is the first-of-its-kind

According to the CDM – Methodological Tool “Tool for the demonstration and assessment of additionality”, version 07.0, this step is optional. Therefore, no analysis or evidence will be provided under this step. The project proponent will prove the additionality starting from Step 1 below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity- Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity.

The project activity is generating electrical energy using renewable energy resources that are fed into the national electricity grid. Hence, the following alternatives are considered:

Alternative 1: The project activity undertaken without being registered under VERRA.

The option of project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Togo national grid, which has existing and new fossil fuel-based power plants that would have been used in the absence of the project activity.

Alternative 2: Continuation of the current situation (No project activity and the equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid)

In such case the project proponent would have withdrawn from an investment in the project activity and continued with its usual business activities. In such case the grid would have continued with the fossil fuel-based power plants resulting in GHG emissions. Hence, existing power plants and new capacity add-ons from fossil fuel-based power plants can be considered as an appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1.a: Identified realistic and credible alternative scenario(s) to the project activity.

Of the two alternatives outlined above, the first alternative is not realistic as project activity as it is not viable without the benefits of carbon credits. The second alternative on the other hand is the baseline scenario for the project activity as per methodology.

Therefore, the continuation of the current situation is the most plausible alternative to the project activity. The project being a green field project activity and the baseline scenario, in line with the methodology, is “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources

Sub-step 1b: The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution.

The implementation of the project activity is consistent with mandatory laws and regulations in Togo and relevant clearances and approvals have been obtained.

Outcome of Sub-step 1.b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the

enforcement in the region or country decisions on national and/or sectoral policies and regulations.

Both the alternatives identified above are found to be in compliance with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and relevant decisions on national and/or sectoral policies and regulations.

However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with the methodology.

Step 2: Investment analysis

An investment analysis is carried out to determine whether the project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of carbon credits. This is demonstrated in line with the tools for the relevant sections as per “Investment Analysis” (ver 9.0).

Sub-step 2a: Determine appropriate analysis method

The project activity envisages the export of renewable energy to the Togo electricity grid and the revenues from the sale of electricity are based on the terms and conditions within the Power Purchase Agreement (PPA).

A simple cost analysis (Option I) cannot be used as the analysis method as the sale of electricity will result in a revenue stream during the project operation. In the absence of the project activity, the continued usage of the grid electricity would have been the most plausible option, which does not require an investment.

Hence an investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option I and Option II, the use of a Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be applied in context of the project activity.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using the Benchmark analysis approach (Option III) has been chosen.

Further, this method illustrates that the assessment of the project by the project proponent was carried out prior to the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality” Version 07.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for the analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for the project activity has been chosen as post tax equity IRR.

According to Methodological Tool27: Investment Analysis Version 9.0, the Default Value Benchmark of Equity IRR in Togo under Group 1, Page 15 (Energy Industries, Energy Distribution, Energy Demand, Waste Handling and Disposal) is 16.25%⁷. Version 9.0 of Tool27 is used as it was valid when the investment decision was taken, which was in March 2019 according to the Memorandum of Understanding.

As the investment analysis is carried out in nominal terms, the benchmark value must also be taken in nominal terms. The nominal benchmark can be calculated using the below formula:

$$\text{Nominal benchmark} = \{(1 + \text{Real benchmark}) \times (1 + \text{Inflation rate})\} - 1$$

Where:

$$\text{Real benchmark} = 16.25\%$$

Inflation rate = 3.2% - As there was no forecasted inflation data available from the Central Bank of West African States (BCEAO) of which Togo is a member, the International Monetary Fund was the reference used. The average forecasted inflation rate for Togo for the first 5 years of the crediting period is 3.2%⁸. Hence, the nominal benchmark value is calculated to be 19.97%

The validation team conclude that, the above-mentioned Benchmark of Equity IRR in Togo is correct and in line with the applied methodology “ACM0002: Grid-connected electricity generation from renewable sources --- Version 20”. Same is also reflected in final VCS PD, version 06. Thus accepted.

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

The Post-Tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e., 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity as shown in the provided spreadsheets.

The IRR and Benchmark analysis shows that the project investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be concluded that the project activity is not the business-as-usual scenario and thus is additional.

<https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v9.0.pdf>

⁸ <https://www.imf.org/en/Countries/TGO#whatsnew>

Post-Tax equity IRR	Benchmark IRR
7.98%	19.97%

The validation team concludes that the above-mentioned Post-Tax equity IRR mentioned under IRR spreadsheet against the Equity IRR in the investment analysis sheet and assumptions considered for the calculation of the same, are appropriate and in line with the applied methodology “ACM0002: Grid-connected electricity generation from renewable sources --- Version 20”. Same is also reflected in final PD version 07. Thus accepted.

Sub-step 2d: Sensitivity Analysis

The objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the robustness of the assumptions used in the development of the investment analysis

The following factor has been subjected to the sensitivity analysis:

1. O&M cost
2. PLF (Plant Load Factor)
3. Tariff Rate

The results of the sensitivity analysis show that even with a variation of +10%/-10% in PLF, O&M costs and Tariff Rate, the Equity IRR is significantly lower than the benchmark IRR. And it is evident from the results given above that the project activity remains additional even under the most favourable conditions.

The validation team confirms that, it is evident that the project is not financially viable even with the carbon benefit. PP accepts this and still has decided to go ahead with the project for environmental and social benefits as confirmed during the interview with PP. Further, PP has chosen to go ahead with the project considering a specific rate of VCU which may increase/decrease is and is beyond the control of PP.

The validation team also confirms that, there are no incentives such as tax holidays provided by the Togo government for project activity.

Sensitivity analysis	Post-Tax Equity IRR			
Variation %	-10%	Normal	+10%	Benchmark
O&M costs	8.59%	7.98%	7.36%	19.50%

PLF	4.69%	7.98%	11.40%	19.50%
Tariff Rate	4.69%	7.98%	11.40%	19.50%

The table below shows the values that breach the benchmark for each of the parameter identified:

Parameter	Original Value	Parameter value that breaches the benchmark	% Difference
O&M (USD Mn.)	1.40	0.00*	-100.00%
PLF @ P50 (%)	20.46%	27.34%	33.59%
Tariff Rate (USD/MWh)	69.49	92.83	33.59%

*Even if the O&M Costs drop by 100%, the equity IRR becomes 14.07% which is still below the benchmark value.

An Investment decision was made by signing the MoU/17/ between AMEA Power and the Togolese Government in March 2019 which triggered hiring consultants and beginning all the studies required to build the plant. Before that the costs was continue to be maintained under Business Development. Same was verified and found consistent by the VVB

The Tariff rate has been considered from the PPA only. The VVB has also checked the annual reports of PP and found there are no changes in tariff rate and is within threshold limit. Thus, the project activity is additional with actual Tariff rate.

Name of the parameter	VVB assessment								
O&M cost and Escalation in the operational expense =2(%)	The details of the proposed project activity are given below. The O&M cost has been considered from O&M contract order and was available at the time decision made for the project activity. The VVB has checked the and found the O&M cost is within threshold limit. Thus, the project activity is additional with O&M cost taken. <table><tr><th>Name of the Investor/Owner/SPVs</th><th>Site Name</th><th>Capacity (MW)</th><th>O&M (In Million) (Without tax)</th></tr><tr><td>AMEA SOLAR TOGO SAU</td><td>Blitta</td><td>70</td><td>1.40</td></tr></table> Based on sectoral scope expert and local knowledge, the project O&M cost and its escalation considered for the proposed project activity is found to be appropriate. Also since the O&M cost is available to VVB and IRR is still within benchmark and thus the same is acceptable.	Name of the Investor/Owner/SPVs	Site Name	Capacity (MW)	O&M (In Million) (Without tax)	AMEA SOLAR TOGO SAU	Blitta	70	1.40
Name of the Investor/Owner/SPVs	Site Name	Capacity (MW)	O&M (In Million) (Without tax)						
AMEA SOLAR TOGO SAU	Blitta	70	1.40						

Tariff	The Tariff rate has been considered from the PPA only. The VVB has also checked the annual reports of PP and found there are no changes in tariff rate and is within threshold limit. Thus, the project activity is additional with actual Tariff rate.			
	Name of the Investor/Owner/SPVs	Project Capacity (MW)	Tariff rate (USD/MWh)	
	AMEA SOLAR TOGO SAU	70	69.49	
PLF	The details of the proposed project activity are given below.			
	Name of the Investor/Owner/SPVs	Site Name	Capacity (MW)	PLF (%)
	AMEA SOLAR TOGO SAU	Blitta	70	20.46%
	Validation team assessed that the PLF is considered as per industry standard. Same standard has been used in the financials and the emission reduction calculation.			
	Assessment team noted that the PP has mentioned that it is the p50 for project activity chosen as PP expects to be part of the project for the whole lifetime (25 years), and therefore, PP is confident that despite the typical yearly fluctuations, after 25 years the total generation will match the cumulative p50 which is crosschecked by the assessment team and found inline and thus acceptable to the assessment team.			

Step 4: Common practice analysis

As per the “Tool for demonstration and assessment of additionality” (Version 07.0.0), Step 2 Investment Analysis shall be complemented with an analysis of the extent to which the project type (e.g., technology or practice) has already diffused in the relevant sector and region within the country. This test is a credibility check to complement the investment analysis. The Methodological Tool followed in this Analysis is Methodological Tool 24: Common Practice, Version 3.1.

Common Practice Analysis:

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity (incl. phase 2)	Unit
+50%	105	MW
Project activity capacity	70	MW
-50%	35	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- the projects are located in the applicable geographical area
- the projects apply the same measure as the proposed project activity
- the projects use the same energy source/fuel and feedstock as the project activity, if a technology switch measure is implemented by the project activity
- the projects implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the project plant
- the capacity or output of the projects is within the applicable capacity or output range calculated in
- the projects started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, whichever is earlier for the proposed project activity.

Analysis of Step 2

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- as the project is located in Togo, projects in the geographical area of Togo have been chosen for the analysis

The validation team noted that as the project is located in Togo, therefore, projects in the geographical area of Togo have been chosen for analysis. Each country has different policies regarding renewable energy, hence considered as geographical region for common practice analysis. The distinction from choosing the Nation to entire geographical boundary is already explained above in the report and thus the applied geographical area is acceptable to the validation team.

- the project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying the same measure (b) are candidates for similar projects

The validation team noted that the project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.

- the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for the analysis

The validation team confirms during the audit that the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.

- the project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects

The validation team confirms during the audit that the project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.

- the capacity range of the projects is within the applicable capacity range from 35 MW to 105 MW

Since the project activity is 70 MW, the output range of +/- 50% has been considered as 105 MW (Higher range for comparison) to 35 MW (Lower range for Comparison) which is assessed to be correct by the validation team.

- as per the methodological tool the projects that have started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, are to be selected for consideration

The start date of the project activity is 12-June-2021. The projects which have started commercial operation till VCS PD preparation date have been considered for analysis.

The findings of the analysis of Step 2

There are no similar projects in Togo that fulfil above-mentioned conditions. BLITTA - TOGO SOLAR POWER PLANT is considered as the first renewable energy facility in the country by International Renewable Energy Agency⁹.

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing project verification.

$N_{all} = 0$

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity.

$N_{diff} = 0$

As per the requirement of step 3, CDM project activities, which have got registered or are under validation, have been excluded in this step. The source of check for the same is cdm.unfccc.int

Step (5): Calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the project activity that delivers the same output or capacity as the project activity.

$F = 1 - N_{diff}/N_{all} = 1 - (0/0) = \text{undefined} \approx 0$

Thus, against the requirements set in the Methodological Tool 24: Common Practice version 3.1, being $F = \text{undefined} \approx 0$ which is less than 0.2 and $N_{all} - N_{diff} = 0$ which is less than 3 the project activity is not a “common practice” within the applicable geographical area.

⁹ <https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility>

The above discussions show that solar power development is not a common practice and the project activity is not financially attractive; hence the project activity is additional. Thus, common practice analysis is acceptable to the validation team.

Conclusion:

The validation team has checked the O&M cost as per the contract and found it is within threshold limit. Based on the project O&M cost and its escalation considered as per O&M contract for the proposed project activity is found to be appropriate for solar power projects. Also, since the O&M cost is available to VVB and IRR is still within benchmark and thus the same is acceptable.

The validation team assessed that the PLF considered and derived from Acceptable Togo Industry Standard for the project activity is within the range of sensitivity analysis and has been defined correctly in a VCS PD, and it is verified by the VVB that it is derived as per the requirements under section 3(a) of Ex-ante definition of the plant load factors of the Guidelines for the reporting and validation of plant load factors, version 01.

The PLF @P50 (%) mentioned in VCS PD is crosschecked by the assessment team with the industry standard and found inline and appropriate.

Hence, it can be concluded by the validation team that the project activity is not a common practice and thus is considered additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

The validation team checked the baseline, project and leakage calculation and confirms that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detailed analysis is as below:

According to para 22 of the applied methodology, ACM0002 Large Scale Consolidated Methodology: Grid-connected Electricity Generation from Renewable Sources, Version 20.0, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system" version 07.0.

The scope and applicability of Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, version 01.0, covers the grid emission factors for the electricity system of the WAPP (West African Power Pool).

This Standardized Baseline can be used under the following conditions:

Baseline Procedure	Justification
The project activity is implemented in any one of the WAPP member countries i.e., Benin, Burkina Faso, Cote d'Ivoire, Ghana, Mali, Niger, Nigeria, Senegal and Togo and is connected to the WAPP	The project activity is in Togo and is connected to the WAPP.
The CDM approved methodology that is applied to the project activity requires the determination of CO ₂ emission factor(s) through the application of the grid tool	ACM0002 requires the determination of CO ₂ emission factor(s) through the application of the grid tool "TOOL07: Tool to calculate the emission factor for an electricity system version 07.0
The project activity uses the ex-ante options for both the operating margin and build margin grid emissions factors, as described in the grid tool, and therefore no monitoring or recalculation of the emission factor during the crediting period is required	The project activity uses the ex-ante options for both the operating margin and build margin grid emission factors, as described in the grid tool hence No-monitoring or recalculation of the emission factor is not required during the crediting period.

According to the ACM0002, Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel-fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above the baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

According to para 4 of the Standardized baseline: Grid emission factor for the West African Power Pool, version 01.0, the latest approved standardized baseline valid value of the CO₂ emission factor is 0.573 tCO₂/MWh applied for the project electricity system since Togo is listed in the WAPP member countries.

The baseline emissions are to be calculated as follows:

$$BE_y = EGP_{J,y} \times EF_{grid,CM,y}$$

Where:

$$BE_y = \text{Baseline emissions in year } y \text{ (tCO}_2\text{/yr)}$$

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" version 07.0 (tCO₂/MWh) and as per the Standardized baseline: Grid emission factor for the West African Power Pool, version 01.0, the Combined margin CO₂ emission factor for the WAPP power system for solar power generation project activities applicable for first crediting period in tCO₂/MWh

Calculation of $EG_{PJ,y}$

In case of Greenfield power plant development, the calculation of the electricity system's combined margin (CM) shall be based on the following equation, according to the methodology:

$EG_{PJ,y}$ = $EG_{facility,y}$ Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

As stated before, the "TOOL07: Tool to calculate the emission factor for an electricity system" allows referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool version 01.0.

The ASB0034-2021 Standardized Baseline states that for wind and solar power generation project activities, the $EF_{grid,CM,y}$ to be used is 0.573 tCO₂/MWh.

Project Emission

The project activity involves renewable power generation. Therefore, $PE_y = 0$. This is found acceptable.

Leakage Emission

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are not calculated being negligible and this is found acceptable by the VVB.

Net GHG Emission Reductions and Removals

Baseline Emissions:

Baseline Emissions are calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂ /year)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y. The value is assumed based on the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using version 01.0 of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh).

Parameters determined ex-ante:

Data / Parameter	EF grid, CM, y
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ grid emission factor in the year y
Source of data	Standardized Baseline ASB0034-2021
Value applied	0.573 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	The availability of a Standardized Baseline for the Grid Emission Factor for West African Power Pool version 01.0
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

Data and Parameters Monitored

Data / Parameter	EG PJ,y
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied to the grid in year y by the solar plant
Source of data	Meter readings

Description of measurement methods and procedures to be applied	Net electricity generation will be electronically metered and stored under the Plant Manager's authority				
Frequency of monitoring/recording	Continuous measurement and monthly recordings				
Value applied	107,027				
Monitoring equipment					
		Meter Description	Serial Number	Accuracy Class	Make
	1	Power Trafo.1 Main	56198213	0.2S	Landis+Gyr
	2	Power Trafo.1 Check	56198214	0.2S	Landis+Gyr
	3	Power Trafo.2 Main	56198215	0.2S	Landis+Gyr
	4	Power Trafo.2 Check	56198216	0.2S	Landis+Gyr
	5	Blitta feeder Main	56198203	0.2S	Landis+Gyr
	6	Blitta feeder Check	56198217	0.2S	Landis+Gyr
	7	Auxiliary Trafo. Main	56198221	0.2S	Landis+Gyr
	8	Auxiliary Trafo. Check	56198222	0.2S	Landis+Gyr
QA/QC procedures to be applied	Calibration of the meters will be carried out as per manufacturer specifications. Inspection and testing of accuracy will be carried out on annual basis				
Purpose of data	Calculation of baseline emissions				

Calculation method	<i>Not applicable</i>
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.

Based on the meter readings and the monthly recordings provided by the PP to the assessment team the quantity of net electricity generated and supplied to the grid is 107,027 MWh/yr by the solar plant.

The total estimated quantity of net electricity supplied to the grid:

$$EG_{pj,y} = 107,027 \text{ MWh/yr}$$

The Grid Emission Factor is:

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The calculation of the baseline emissions of the project activity is:

$$BE_y = 107,027 \text{ MWh/yr} * 0.573 \text{ tCO}_2/\text{MWh} = 61,326 \text{ tCO}_2/\text{yr}.$$

Leakage:

Leakage emissions are not applicable.

Project Emissions:

Project Emissions are as shown below:

$$PE_y = 0 \text{ tCO}_2/\text{yr}$$

Emission Reductions:

$$ER_y = BE_y - PE_y$$

$$ER_y = 61,326 \text{ tCO}_2/\text{yr} - 0 \text{ tCO}_2/\text{yr} = 61,326 \text{ tCO}_2/\text{yr}$$

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
Year 2021	39,985	0	0	39,985
Year 2022	70,822	0	0	70,822
Year 2023	70,468	0	0	70,468

Year 2024	70,308	0	0	70,308
Year 2025	69,765	0	0	69,765
Year 2026	69,416	0	0	69,416
Year 2027	69,069	0	0	69,069
Year 2028	30,774	0	0	30,774
Total	490,607	0	0	490,607

3.3.7 Methodology Deviations

The validation team confirms that no methodology deviation is applied to the project activity.

3.3.8 Monitoring Plan

The proposed Monitoring Plan describes the Management System and Procedures to be developed and implemented by the project owner for this project in order to ensure consistency with the methodology ACM0002: Grid-connected electricity generation from renewable sources - -- Version 20.0 which is applied in the context of the project activity.

The developed monitoring plan contains the following sections:

1. Monitoring parameters
2. Monitoring Points
3. Monitoring equipment
4. Management structure
5. Monitoring guidelines, procedures and documentation
6. Regular staff training

All data required for the calculation of emission reductions associated with the project activity will be monitored in accordance with the approved methodology ACM0002: Grid-connected electricity generation from renewable sources -- Version 20.0 and will be archived electronically and on paper throughout the crediting period and up to 2 years thereafter.

A Monitoring Management and Quality Assurance System (MMQAS) will be developed and implemented to ensure that the results from the project activity are reliable and transparent.

The complete MMQAS, including data and documents from project operation, will be open for review to the VVBs contracted for verification.

Monitoring Parameters

The tables under section 5.1 provide information on the required monitoring parameters as per approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 and referred tools, required for the ex-post calculation of project emissions associated with the project activity.

Monitoring Points

Are defined as per the single line diagram provided to the VVB and points for monitoring summarized in Figure 2 in the PD version 07.

Monitoring equipment

The physical monitoring system consists of the following meter equipment:

Electricity meter: AMEA Solar TOGO SAU to monitor the quantity of net electricity generation supplied to the grid in year y by the solar plant ($EG_{PJ,y}$)

As shown in the calculations of section 4.4 of the PD version 07 and according to the methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20., baseline emissions will be calculated by multiplying the quantity of net electricity generation supplied to the grid in year y by the solar plant ($EG_{PJ,y}$ with) with the grid emission factor in the year y ($EF_{grid, CM, y}$)

Management structure

A VCS team will be designated to be responsible for carrying out the required monitoring tasks for the project activity. Both management people and technical workers will be present in the VCS team. All of the team members must have passed internal monitoring training.

The management structure consists of 3 levels to guarantee a step-by-step management of the project activity, which will cast concrete responsibilities on the appointed person who will be actually in charge of a specific task.

The 3 levels refer to:

1. Solar Plant Manager
2. VCS project manager
3. VCS staff

Before the start of the crediting period, clear roles and responsibilities will be assigned to all staff involved in the project activity. A VCS Project Manager will be appointed who will be responsible for monitoring activities and data management associated with the project activity. All operators involved in the collection of data and records will be coordinated by the VCS Manager.

Monitoring guidelines, procedures and documentation

The introduced Monitoring Management and Quality Assurance System is building on written guidelines and procedures provided to the operational VCS team staff and to the local VCS management.

The procedures will be developed by the external consultant according to the approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0. All tasks under the monitoring plan will be carried out strictly according to the provided guidelines and procedures.

The Monitoring Management and Quality Assurance System documentation is designed to enable VCS staff members to perform the required tasks following the given guidelines and procedures. Providing documentation designed specifically for the required tasks will strongly benefit availability and transparency of data streams and processes which are relevant for the VCS project.

The VCS Project Manager will carry out frequent performance and data cross-checking, to ensure that staff are following the Monitoring Management and Quality Assurance System.

All documents filled out and data acquired during the project will be archived separately for backup.

Regular Staff Training

The Monitoring Management and Quality Assurance System is implemented and will be maintained by an external consultant. In order to qualify local operational staff to carry out required regular tasks external consultant will provide professional training, both in class and practical application, to the designated VCS team staff before the start of the crediting period and ongoing regular training at least once a year for the duration of the project.

Only trained staff will be members of the VCS team and only trained VCS staff members will be allowed to perform any tasks related to monitoring activities.

The VCS Manager is permitted to train new staff. By this it will be ensured that operational VCS staff are qualified to carry out the relevant tasks under the Monitoring Management and Quality Assurance System.

Based on the annual monitoring results VCS staff will prepare the annual Monitoring Report. An overall AQ/QC on the Monitoring Spreadsheet and Monitoring Report will be carried out by an external Consultant before the documents are submitted to a verifying VVB.

The validation team checked the monitoring practice on site and checked the same with the defined monitoring plan as per VCS PD, The VVB is of the opinion that, the project activity is being monitored adhering to the monitoring plan to the requirements of the applied methodology and referred tools.

3.4 Non-Permanence Risk Analysis

Not applicable for the project activity.

4 VALIDATION CONCLUSION

LGA Technological Center, S.A. (Applus+ Certification) has been engaged by “ClimatePartner GmbH” to perform the validation of the project activity “BLITTA - TOGO SOLAR POWER PLANT”

The management of the project participant/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the VCS PD and the approved methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 20.

Our Validation approach was based on the requirements as defined by the VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these.

The validation team confirms that:

- The projects description compliance with, the requirements and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline V4.2 and standard version 4.3.
- The project’s baseline and additionality are assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 20 and Tool for the demonstration and assessment of additionality version 07.0,
- The project’s monitoring plan is assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 20
- The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGA Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.
- The project is expected to generate 490,607 tCO_{2e} during the length of entire crediting period (12-June-2021 to 11-June-2028).

APPENDIX 1: DOCUMENTS REVIEW UNDER VALIDATION

No.	Author	Title	References to the document	Provider
1.	AMEA SOLAR TOGO SAU	Commissioning certificates	Commissioning certificates of the Solar Panels implemented in the project site, Declaration of Conformity, AMEA TOGO SOLAR S.A. AMEA TECHNICAL SERVICES LTD-28102022180209 (003) Executed version.pdf - Page 167	Project participant
2.	N/A	Ownership Documents	1. Certificate of Completion signed by the Togolese Republic 2. Power Purchase Agreement (PPA) signed by AMEA SOLAR TOGO SAU and the Togolese Republic 3. Declaration of Conformity signed by the Togolese Republic 4. Entry into Force of the Concession Agreement signed by AMEA SOLAR TOGO SAU and the Togolese Republic 5. Back Energization of the PV Plant Document signed by the Togolese Republic, the Electricity Off-taker and AMEA SOLAR TOGO SAU	Project participant
3.	Jinko Solar, Sungrow	O&M Word file	1. Monitoring manual of the project site 2. O&M contract information with the manufacturer or the engineering firm	Project participant
4.	Jinko Solar	SLD File	Single Line Diagram	Project Participant
5.	Applus	Contract of the PP with the VVB	Contract of the PP with the VVB	Project participant
6.	Jinko Solar, Sungrow,	Technical specifications	Technical specifications of each SPV from manufacturers	Project participant

No.	Author	Title	References to the document	Provider
7.	PP	Initial VCS PD version 01 Final VCS PD version 07	17-May-2022 11-April-2023	Project participant
8.	PP	Emission Calculation sheet- version 01 Emission Calculation sheet- version 07	17-May-2022 23- March-2023	Project participant
9.	Jinko Solar, Sungrow,	The operational lifetime of the project activity from the manufacturer= (Technical specifications)	Manufacturer technical specifications	Project participant
10.	NA	Reference link is provided.	https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility https://www.mofaic.gov.ae/en/mediahub/news/2021/6/24/24-06-2021-uae-togo https://data.worldbank.org/indicator/EG.E.LC.ACCS.ZS?locations=TG https://unfccc.int/sites/default/files/NDC/2022-06/CDN%20Revis%C3%A9es_Togo_Document%20int%C3%A9rimaire_rv_11%2010%2021.pdf https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility	Independent Search

No.	Author	Title	References to the document	Provider
11.	UNFCCC VERRA	Tools/ guidelines used in the project activity	UNFCCC CDM web site • UNFCCC Methodology: ACM0002: Grid-connected electricity generation from renewable sources — Version 20 • Tool: Tool for the demonstration and assessment of additionality version 07.0 • VCS Validation Report Template Version 4.1 • Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using version 01.0 • TOOL01: Tool for the demonstration and assessment of additionality version 07.0	UNFCCC
12.	PP	VCS Declaration	Declaration from PP for Participation under Other GHG Programs	Project participant
13.	PP	LSC	Local stakeholders' consultation supporting; Notices, MoM's & photographs etc.	Project participant
14.	PP	PPA	Power purchase agreement	Project participant
15.	PP	VCS Declaration	no ODA declaration	Project participant
16.	PP	EIA	ESIA and ESMP	Project participant
17.	PP	MoU	PROTOCOLE D'ACCORD	Project participant

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

CAR ID	01	Section no.	4.1	Date: 17-August-2022
Description of CAR				
During review of monitoring report following inconsistencies observed:				
1. PP requested to review and resubmit VCS PD template version 4.1 with applying requirements of latest VCS Standard, v4.3. Thus, corrective action sought. 2. Date format found not inline with VCS PD template requirements. Thus, corrective action sought. 3. PP have mentioned the start date of the project activity, however commissioning dates individual/batchwise, found missing in the VCS PD. 4. In sec. 1.1, PP shall provide supporting references/ weblinks to support the statement that this project is the largest in west Africa and first renewable energy facility in Togo 5. PP shall provide copies of documents referred in sec. 1.7 of the PDD 6. In sec. 1.17 , PP shall provide supporting documents/ references to justify the statements regarding current status of electricity access in the host country.				
Project participant response				Date: 28-October-2022
1. The PD has been updated complying to the requirements of the latest v4.3 VCS Standard. 2. The date format has been updated to DD-Month-YYYY according to the template requirements. 3. Commissioning date has been added to sec. 1.8 along with the project activity start date. 4. References to support the statement have been provided in the PD. The references are sourced from reliable websites including International Renewable Energy Agency (IRENA). 5. Copies of documents referred in sec. 1.7 of the PD have been provided in the folder 'CAR ID 01'. 6. References from reliable sources provided to support the statement. The references have been sourced from World Bank Data and the Nationally Determined Contribution (NDC) of Togo from the UNFCCC Database.				
Documentation provided by project participant				
Certificat sur Honneur.pdf CAE_Avenant au Contrat d'achat d'énergie_signé_scanné.pdf Declaration of Plant Conformity.pdf PV de mise en vigueur de la convention de concession.pdf PV mise sous tension.pdf				
VVB assessment				Date: 28-October-2022
1. PP shall review and resubmit VCS PD template version 4.1 with applying requirements of latest VCS Standard, v4.3, hence this part of CAR is open. 2. Date format is now found in line with VCS PD template requirements; hence this part of CAR is closed. 3. PP has mentioned the start date of the project activity, however commissioning dates individual/ proposed batch wise, found missing in the VCS PD, hence this part of CAR is open. 4. PP has now provided the appropriate weblink which was cross-checked and confirmed by the assessment team, hence this part of CAR is closed. 5. It is cross-checked and confirmed from the Certificate on Honor provided by the Togolese Republic, represented by the Minister of Economy and Finance and the Minister Delegate to the President of the Republic in charge of Energy and Mines to the PP, by the assessment team, hence this part of CAR is closed. 6. The reference provided by the PP are cross-checked and confirmed by the assessment team, hence this part of CAR is closed.				
CAR is open				
Project participant response				Date: 14-November-2022

1. The PD has been updated complying to the requirements of the latest v4.3 VCS Standard.	
3. The commissioning dates of the phases 1 and 2 have been added to the VCS PD.	
Documentation provided by project participant	
VVB assessment	Date: 14-November-2022
1. PP has now updated the VCS PD as per the VCS PD template 4.1 incorporating the latest VCS Standard V4.3, hence this part of CAR is closed.	
2. PP now has mentioned the Phase wise commissioning dates of the project hence this part of CAR is closed.	
CAR is closed	

CAR ID	02	Section no.	4.1	Date: 17-August-2022
Description of CAR				
1- PP requested to review & justify the methodological tool 23 of the applied methodology ACM0002 Version 20.0. Corrective action sought.				
2- PP shall provide no ODA declaration and an undertaking that there will be no double counting of emission reductions				
Project participant response				Date: 28-October-2022
1. The reference to the methodological tool 23 has been removed as it has not been used to prove the additionality.				
2. A No ODA declaration is provided with an undertaking that there will be no double counting of emission reductions in the folder 'CAR ID 02'.				
Documentation provided by project participant				
No ODA Togo Blitta_Signed.pdf				
VVB assessment				Date: 28-October-2022
1. PP has now updated the VCS PD, hence this part of CAR is closed.				
2. PP has now provided the no ODA declaration which was cross-checked and confirmed by the assessment team and found inline, however, there will be no double counting of emission reductions undertaking was not found in the folder CAR ID 02 provided by the PP, hence this part of CAR is open.				
CAR is open				
Project participant response				Date : 14-November-2022
2.No double counting of emission reductions undertaking is provided in the folder 'CAR ID 02' in 'Validator Comments - Supporting documents.'				
Documentation provided by project participant				
Togo SPV Undertaking v 2_Carbon Credits Registration_2022 11 02 clean.docx				
VVB assessment				Date: 14-November-2022
PP has provided the document, undertaking letter dated 3 November 2022 stating and confirming that, there will be no double counting of emission reductions which was reviewed by the assessment team and hence this part of CAR is closed.				
CAR is closed				

CAR ID	03	Section no.	4.1	Date: 17-August-2022
Description of CAR				
1. Update section 2.4				
2. PP shall provide copy of ESIA and ESMP				
Project participant response				Date: 28-October-2022
1. Section 2.4 has been updated upon the closure of the public comment period.				
2. Copy of ESIA and ESMP have been provided in the folder 'CAR ID 03'. As ESMP is added as a part of the ESIA, there shall only be one document with both.				
Documentation provided by project participant				
DNVGL_Togo_30MW Solar PV_ESIA_23072019_English.pdf				
DNVGL_Togo_50MW Solar PV_Addendum 17052020.pdf				
VVB assessment				Date: 28-October-2022

1. PP has now updated section 2.4, hence this part of CAR is closed. 2. PP has now provided the copies of ESIA which was cross-checked and confirmed by the assessment team, hence this part of CAR is closed.
CAR is closed.

CAR ID	04	Section no.	4.2	Date: 17-August-2022
Description of CAR				
1. PP shall provide links/ documents referred for all the input parameters for investment analysis in the tab “assumption” 2. As project is commissioned already, PP shall provide copies of purchase orders, O&M Contract, Board resolution to substantiate the date of investment decision, Loan sanction letter if applicable. 3. PP shall also provide references/ web links used for common practice analysis.				
Project participant response				Date: 28-October-2022

1. The updated investment analysis with links/documents referred is provided in the folder 'CAR ID 04' with references in the folders '02. Financial Evidence +Excerpts', '03. O&M Contract ' and '05. Tariff PPA Excerpt'.

AMEA SOLAR TOGO SAU has arranged project financing for this project with 75% contribution from External Debt and 25% contribution from AMEA Equity. This Debt/Equity ratio is maintained throughout the construction and operation as stated in the Article 16 of the Retrocession Agreement signed with the Togolese State.

15. ENGAGEMENTS FINANCIERS

Les engagements stipulés au présent Article 15 (*Engagements Financiers*) entrent en vigueur à compter de la Date de Signature, et le demeureront tant qu'un montant quelconque restera dû au titre des Documents de Financement.

15.1 Niveau Minimum de Fonds Propres

15.1.1 Le Ratio de Fonds Propres devra être supérieur ou égal à 25% (vingt-cinq pour cent) à la Date de Mise en Service de la Centrale ou, à défaut, au niveau agréé entre l'Emprunteur et le Prêteur et indiqué dans le Modèle Financier tel que mis à jour à la Date de Mise en Service de la Centrale dans les conditions prévues au paragraphe 14.3 (*Modèle Financier*), lequel sera au moins égal au niveau minimum requis pour que les obligations de l'Emprunteur au titre du paragraphe 15.2.1 ci-après soient respectées.

15.1.2 Si le montant des Fonds Propres mis à disposition est insuffisant pour permettre le respect de ce ratio, le Sponsor devra mettre à la disposition de l'Emprunteur des Fonds Propres additionnels (le cas échéant sous forme de nouvelles avances en compte courant, dans le respect des stipulations de la Convention de Subordination), en quantité suffisante, au plus tard à la Date de Mise en Service de la Centrale.

Translation:

15. Financial commitments

The commitments stipulated in this Article 15 (financial commitments) come into force from the Date of Signature, and will remain so as long as any amount remains due under the Finance Documents.

15.1 Minimum Level of Own Funds

15.1.1 The Equity Ratio must be greater than or equal to 25% on the Plant Commissioning Date or, failing that, at the level agreed between the Borrower and the Lender and indicated in the Financial Model as updated at the Date of Commissioning of the Plant under the conditions provided for in paragraph 14.3 (Financial Model), which will be at least equal to the minimum level required for the Borrower's obligations under paragraph 15.2.1 below to be respected

15.1.2 If the amount of Own Funds made available is insufficient to enable compliance with this ratio, the Sponsor shall make available to the Borrower additional Own Funds (if applicable in the form of new current account advances, in compliance of the stipulations of the Subordination Agreement), in sufficient quantity, at the latest on the Date of Commissioning of the Plant

Accordingly, AMEA contributed its Equity to AMEA Togo Solar SAU (Project Company) in the form of share capital to meet local requirement compliance, and in the form of shareholder loan for flexibility towards injection and repayment during the project lifetime.

The above has been booked and recognized in the Audited Financial Statements of Project Company, as share capital in Note 13 "CA - Capital" and as shareholder loan in Note 17 "DJ - Fournisseurs d'exploitation" (refer to Page 7 as Balance Sheet). The audited financials and shareholder loan can be found in '06. Equity Contributions' in 'CAR ID 04'.

The Note 17 refers to the balance payable towards related and unrelated parties. The treatment of the shareholder loan as payable is compliant with the OHADA regulation as advised by our statutory auditors in Togo (Deloitte), and group auditors (EY).

2. The copies of purchase orders, O&M Contract, Board resolution have been provided in the folder 'CAR ID 04'. In regard to the O&M evidence and pricing, you'll notice a discrepancy between the O&M contract (which is in EUR) and the O&M consideration in the IRR calculation. This is because in addition to the O&M contract, there are additional OPEX considerations throughout the lifetime of the Plant including:

- Maintenance Reserve Account (MRA) for capital spares and unscheduled maintenance which we don't have quotes for as this is a budget for "unexpected" expenses throughout the Plants lifetime.
- SPV costs including; personnel hired under the project company, office expenses, third party local advisors e.g. auditors and legal counsel, travel costs, permits and license annual fee's associated to the operation of the Plant.
- Back-office costs including in-house personnel supporting the operations of the project such as the engineering team, asset management, legal team, accounting and finance.
- Annual budget for CSR activities for which the team has flexibility to allocate as needed

3. A web link has been referenced in the PD to prove that the project activity is not a common practice. Please refer to section 3.5, Page 31 in the PD.

Documentation provided by project participant

220727_Blitta Solar Investment analysis_V2.xlsx All purchase orders are listed in the folder '04. Purchase Orders' Board Resolution Meeting - Investment Decision (FR).pdf O&M SoW - Contract Excerpt.pdf	
VVB assessment	Date: 28-October-2022
1. PP shall provide links/ documents referred appropriately for all the input parameters for investment analysis in the tab assumption, hence this part of CAR is open. 2. PP has now provided copies of purchase orders, O&M Contract, Board resolution to substantiate the date of investment decision, as well as the loan sanction letter, which are cross-checked by the assessment team and found in line so as to substantiate the date of investment decision, hence this part of CAR is closed. 3. PP has now provided web link used for common practice analysis which was cross-checked by the assessment team and found in line with the common practice analysis in project activity requirement, hence this part of CAR is closed. CAR is open	
Project participant response	Date: 14-November-2022
1.The links/ documents referred appropriately for all the input parameters for investment analysis have been provided.	
Documentation provided by project participant	
221107_Blitta Solar Investment analysis_V2.xlsx	
VVB assessment	Date: 14-November-2022
PP has now provided proper links/ documents referring to all the input parameters for investment analysis in the tab assumption, hence this part of CAR is closed. CAR is closed	

CAR ID	05	Section no.	4.4	Date: 17-August-2022
Description of CAR				
PP requested to submit estimated Emission Reduction spread sheet.				
Project participant response				Date: 28-October-2022
Estimated Emission Reduction spread sheet has been provided in the folder 'CAR ID 05'.				
Documentation provided by project participant				
Togo-Estimated Emission Reduction.xlsx				
VVB assessment				Date: 28-October-2022
PP shall crosscheck and resubmit the updated emission reduction sheet to the assessment team, hence this part of CAR is open. CAR is open				
Project participant response				Date: 14-November-2022
The emission reduction sheet is updated and provided to the assessment team in the folder 'CAR ID 05' in 'Validator Comments- Supporting Documents'.				
Documentation provided by project participant				
221107_Togo_Estimated Emission Reduction_commented.xlsx				
VVB assessment				Date: 14-November-2022
PP shall crosscheck and resubmit the updated emission reduction sheet and VCS PD to the assessment team, including year-wise emission reduction calculation mentioning parameters such as Generation Capacity (MWh), No. of Hours Operation Annually (hrs), Capacity Utilization Factor (%), Annual degradation (%), Net Electricity Exported to Grid (MWh), till then this part of CAR is open. CAR is open				
Project participant response				Date: 17-November-2022
The emission reduction sheet is updated and provided to the assessment team in the folder '221117_Responses_DVR Round 3_Togo_VCS'.				
Documentation provided by project participant				
221116_Togo_Estimated Emission Reduction.xlsx				
VVB assessment				Date: 17-November-2022

PP has now resubmitted the updated emission reduction sheet and VCS PD to the assessment team, including year-wise emission reduction calculation mentioning parameters such as Generation Capacity (MWh), No. of Hours Operation Annually (hrs), Capacity Utilization Factor (%), Annual degradation (%), Net Electricity Exported to Grid (MWh), hence this car is closed.

CAR is closed

CAR ID	06	Section no.	4.5	Date: 17-August-2022
Description of CAR				
1. VCS PD does not have information of metering equipment and calibration frequency along with monitoring arrangements at site inline with requirements of applied methodology. Thus, PP requested to revise and resubmit VCS PD.				
2. PP shall provide training records				
Project participant response				Date: 28-October-2022
1. Documents with information on the metering equipment and calibration frequency along with monitoring arrangements at site inline with requirements of applied methodology are provided.				
2. Training records are provided.				
Documentation provided by project participant				
The information on the metering equipment and calibration frequency are provided in the folder '01. Energy Metering' in 'CAR ID 06'.				
All training records are listed in the folder '02. Training Records' in 'CAR ID 06'				
VVB assessment				Date: 28-October-2022
1. VCS PD now contains the information of metering equipment and calibration frequency along with the monitoring arrangements at site in line with the requirements of applied methodology, hence this part of CAR is closed.				
2. PP shall check the documents provided in the folder CAR ID 06 and provide training records, hence this part CAR is open.				
CAR is open				
Project participant response				Date: 14-November-2022
2.Training record supporting documents are provided in the folder 'CAR ID 06' in 'Validator Comments -Supporting documents'				
Documentation provided by project participant				
TrainingRecords.pdf				
1.TOGOLESE Training Programme.xlsx				
VVB assessment				Date: 14-November-2022
PP has now provided evidences and records for the trainings conducted at site, hence this part of CAR is closed.				
CAR is closed				

CAR ID	07	Section no.	4.1	Date: 27-December-2022
Description of CAR				
Reference for location not found in VCS PD, Corrective action is sought.				
Project participant response				Date: 27-December-2022
Reference document is shared				
Documentation provided by project participant				
Blitta PV Plant.kmz				
VVB assessment				Date: 27-December-2022
The reference document shared by PP is found in line, and the location of the project activity is now confirmed, hence this part of CAR is closed.				
CAR is closed				

CAR ID	08	Section no.	4.1	Date: 27-December-2022
Description of CAR				

SDG Goal: Good Health and Wellbeing: No 3's Monitoring Parameter says "Malaria incidence per 1,000 population" will be the parameter to ensure that the "Ensure healthy lives and promote well-being for all at all ages" the Value for the said parameter is not mentioned in this table, Corrective action is sought. PP to justify the same by providing a reference document/link to substantiate the statement "Ensure healthy lives and promote well-being for all at all ages" with respect to the value considered.	
Project participant response	Date: 27-December-2022
Updated VCS PD is now shared	
Documentation provided by project participant	
Updated VCS PD	
VB assessment	Date: 27-December-2022
PP has now updated the table under 1.17: Sustainable Development Contributions mentioning all the monitoring parameters appropriately, hence this CAR is closed CAR is closed	

CAR ID	09	Section no.	4.1	Date: 27-December-2022
Description of CAR				
Provide a demonstration of the applicability conditions of each tool used, apart from the applicability conditions of the methodology applied. Consider Standardized baseline applicability conditions if not yet, corrective action is sought.				
Project participant response				Date: 27-December-2022
Updated VCS PD now incorporates a demonstration of all the applicability conditions of each tool used, apart from the applicability conditions of the methodology applied. Standardized baseline applicability conditions are also Considered.				
Documentation provided by project participant				
Updated VCS PD				
VB assessment				Date: 27-December-2022
PP has now submitted the updated VCS PD now incorporating a demonstration of all the applicability conditions of each tool used, apart from the applicability conditions of the methodology applied standardized baseline applicability conditions are also considered, and found in line, hence this CAR is closed. CAR is closed				

CAR ID	10	Section no.	4.2	Date: 27-December-2022
Description of CAR				
Appropriate values and formula for deriving/ considering the following parameter not found in the Excel sheet, corrective action is sought. On Page: Assumption 1. Value mentioned for Benchmark (Equity IRR) in cell no E3. On Page: IRR 1. % operations - Phase 1 for year 2021 2. % operations - Phase 2 for year 2023 3. In the formula applied for Annual Generation for all the years what is the source of value 50				
Project participant response				Date: 27-December-2022
Appropriate values and formula for deriving/ considering the following parameter are now mentioned in the Excel sheet. On Page: Assumption 1. Value mentioned for Benchmark (Equity IRR) in cell no E3. On Page: IRR 1. % operations - Phase 1 for year 2021 2. % operations - Phase 2 for year 2023 3. In the formula applied for Annual Generation for all the years what is the source of value 50				
Documentation provided by project participant				
Updated 220727_Blitta Solar Investment analysis_V2_Reviewed_SM_0612022				
VB assessment				Date: 27-December-2022
PP has now submitted the updated excel sheet incorporating the appropriate values, formulas and referencing for the said parameters, and found in line by the assessment team, hence this part of CAR is closed. CAR is closed				

CAR ID	11	Section no.	4.2	Date: 27-December-2022
Description of CAR				
“From the IRR sheet, it is evident the project is not financially viable even with the carbon benefit.”, Justify				
Project participant response				Date: 27-December-2022
Yes, it is evident that the project is not financially viable even with the carbon benefit. PP accepts this and still has decided to go ahead with the project for environmental and social benefits as confirmed during the interview with PP. Further, PP has chosen to go ahead with the project considering a specific rate of CER which may increase/decrease is and is beyond the control of PP				
Documentation provided by project participant				
Updated VCS PD				
VVB assessment				Date: 27-December-2022
PP has now provided justification for going ahead with the project considering a specific rate of CER which may increase/decrease and is beyond the control of PP, which is found appropriate and in line to the assessment team, hence this part of CAR is closed. CAR is closed				

CAR ID	12	Section no.	4.2	Date: 27-December-2022
Description of CAR				
In the sensitivity analysis, the value for each of the evaluated parameters that will make the benchmark to be breached not found corrective action is sought.				
Project participant response				Date: 27-December-2022
In the sensitivity analysis, the value for each of the evaluated parameters that will make the benchmark to be breached is now incorporated.				
Documentation provided by project participant				
Updated VCS PD				
VVB assessment				Date: 27-December-2022
PP has now incorporated the evaluated parameters that will make the benchmark breach in the sensitivity analysis, hence this CAR is closed CAR is closed				

CAR ID	13	Section no.	4.2	Date: 27-December-2022
Description of CAR				
“It was observed that investors prefer P90 value while considering the PLF. https://blog.greensolver.net/en/p50-p90-simplified-two-figures-all-investors-should-understand-to-build-a-reliable-business-plan-when-investing-in-wind-assets/ ”, justify				
Project participant response				Date: 27-December-2022
In the experience of the team (more than 10 GW of renewable energy projects across the five continents) the financial metrics used are based on: 1. p90 and DSCR of 1.25x in the case of lenders. This means that lenders take the p90 generation, apply a discount of 25% on the cash flow after taxes, and ensure that there is enough cash flow for debt repayment 2. p50 for project developers with long-term investment profile like AMEA SOLAR TOGO SAU. In our case, we expect to be part of the project for the whole lifetime (25 years), and therefore, we are confident that despite the typical yearly fluctuations, after 25 years the total generation will match the cumulative p50. There are some slight differences to the above. In example, a project developer with short-term appetite might look to exit after 2 or 3 years and therefore will not look at the p50 or p90 but rather the financing structure, since it will have a bigger impact than a +/- 7% production difference in 2 years. Furthermore, note that as the project is constructed and operational, the risks decrease and some lenders with more risk appetite look at other percentiles when re-financing (i.e. p75) The above is worldwide industry standards, there are not Togo standards				
Documentation provided by project participant				
Updated VCS PD, Assumption sheet				
VVB assessment				Date: 27-December-2022

PP has mentioned that it is the p50 for project activity chosen as PP expects to be part of the project for the whole lifetime (25 years), and therefore, PP is confident that despite the typical yearly fluctuations, after 25 years the total generation will match the cumulative p50 which is crosschecked by the assessment team and found inline hence this CAR is closed.
CAR is closed

CAR ID	14	Section no.	4.4	Date: 27-December-2022
Description of CAR				
Deration factor to be applied every year in crediting period and PP to apply the same to pervious years Annual Generation (MWh/Year) appropriately in ER sheet and mention the same values in PD, corrective action is sought.				
Project participant response				Date: 27-December-2022
Deration factor is applied to the entire crediting period in ER sheet and mentioned the same values in PD.				
Documentation provided by project participant				
<i>Updated VCS PD</i>				
VVB assessment				Date: 27-December-2022
PP has now applied the deration factor to the entire crediting period appropriately in ER sheet and mentioned the same values in PD, hence this CAR is closed CAR is closed				

APPENDIX 3: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Mr. Kumar	Pankaj	Mr. Pankaj Kumar has done M. Sc in Environment Management from Forest Research Institute, Dehradun and B. Sc. (Hons.) in Environment & Water Management from Magadh University, Bihar, India. He has also done Post Graduate Diploma in Environmental Law from NLSIU, Bangalore. He has more than 12 years of working experience in GHG Assessments and has participated during his career in Agencies and DOEs like MITCON, Agrinergy, Carbon Check and is empaneled with Applus+ Certification since 2015 for the performance of CDM/VCS/GS project assessments. He has extensive experience in the Renewable, Waste Management and Energy Demand Scopes of UNFCCC CDM and has done more than 100 Validations and Verifications of PAs and PoAs as Lead Auditor, Technical Expert and Technical Reviewer, mainly in Asia, Africa, USA, Asia Pacific and Americas under CDM, Verified Carbon Standard, Gold Standard & Social Carbon Standard, Brazil. He is an experienced, qualified and result oriented Environment and climate change professional having 16 yrs. of relevant experience in Climate Change (Mitigation & Adaptation), Environmental Due Diligence, Disaster Risk Reduction, Climate finance, adaptation planning, capacity building, validation and verification of GHG project. He can also provide technical support for environmental investigative, remedial projects involving air, water and soil, Waste management, EIA, Environmental Compliance, ISO 14001, OHSAS 18001, GHG accounting (ISO 14064) and Carbon foot printing.
Mr. Meesa	Srikanth	Mr. Srikanth Meesa is a climate change professional with more than 16 years of experience in the fields of climate change, GHG auditing, carbon footprint assessments, water and energy audits, scope-3 emissions, net zero strategies, sustainability & ESG sectors and water and wastewater treatment sectors. He has completed his Master's in Environmental Engineering& Management from I.I.T Delhi and his Bachelor's in Civil Engineering from Osmania University. Furthermore, he has extensive experience i.e., more than 14 years in the auditing of CDM, VERRA, GS, JCM projects of various sectoral scopes across the world. He has worked with reputed certification bodies such as TUV NORD, DNVGL, TUV SUD, and LR. He also has consulting experience on sustainability reports& road-maps, stakeholder consultation, and LCA studies while working at Thinkstep (Now called Sphera). Currently, he is associated with Global Green Solutionz and empaneled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Mr. Srikanth Meesa is based at Hyderabad, India.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification request
CM	Combined Margin
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DOE	Designated Operational Entity
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GWP	Global Warming potential
VVB	Validation/Verification Body

APPENDIX 5: STAKEHOLDERS CONSULTATION

A Local Stakeholder Consultation process has been conducted in accordance with the VERRA- Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity.

The Local Stakeholder Consultation process took place as follows:

The Local Stakeholder Consultation process was held on Tuesday, 05-July-2022 at CICB (Centre International de Conférence de Blitta), and attended by Prefect (Village Chief), local persons including local villagers, local vendors, and technology suppliers. All the stakeholders were invited through an invitation letter dated 20-June-2022 to attend the stakeholders meeting.

The stakeholders identified by the project participant were local villagers who are the major population of the area, local communities, and Prefect (Village Chief), EPC Contractor, project proponent representatives, O&M Team and other people involved in the project. Validation team verified the list of participants who attended the stakeholder meeting and feedback questionnaire and confirms the stakeholders identified are relevant.

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The validation team also verified the minutes of meeting to note that no negative comments were received and the same was cross-checked with the information obtained during follow-up interviews with the stakeholders.

Thus, VVB concludes that, the stakeholder's consultation was conducted in accordance with the VERRA- Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity.